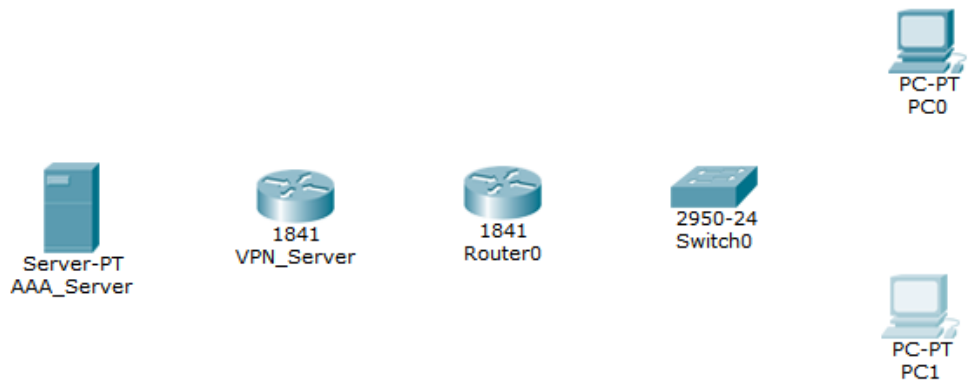
Your task is to create a Virtual Private Network (VPN) using Authentication, authorization and accounting (AAA). AAA is a system for tracking user activities on an IP-based **network** and controlling their access to **network** resources. **AAA** is often implemented as a dedicated server. This term is also referred to as the **AAA** Protocol.

Open the packet labelled VPN.pkt on the student common

You must use PacketTracer v6.2

Save the packet as SurnameFirstname.pkt (remember to save the packet after each step)
Save this word document as SurnameFirstname. (remember to save after each step)
You must submit both the packet and the word document in one folder labelled SurnameFirstname

Step 1:



Add the following devices as shown in the above diagram. The VPN_Server router has been added to the packet already

Server generic
Router 1841
Switch 2950-24
PC x2

Label the server-PT AAA_Server as shown

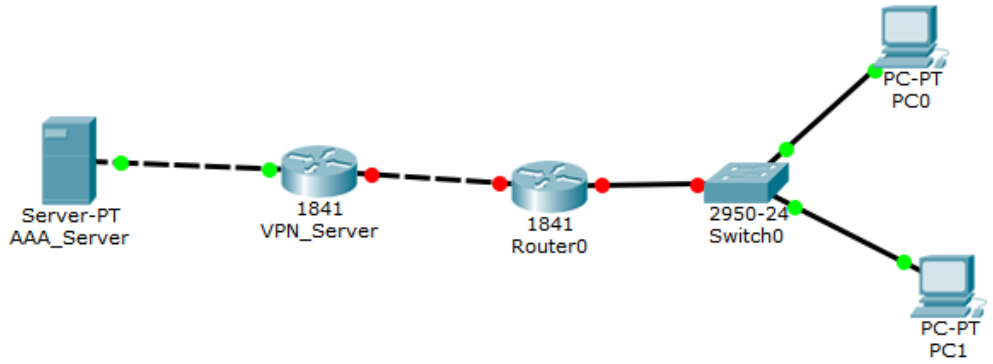
Step 2:

Connect the devices.

Using automatically select the cables for the connection

N.B (you must connect the devices in the following order)
1. Connect the 2 routers first
2. Connect the router VPN_Server to the server
3. Connect router 1 to the switch
4. Connect PC0 to the switch
5. Connect PC1 to the switch

Your network should now look as below



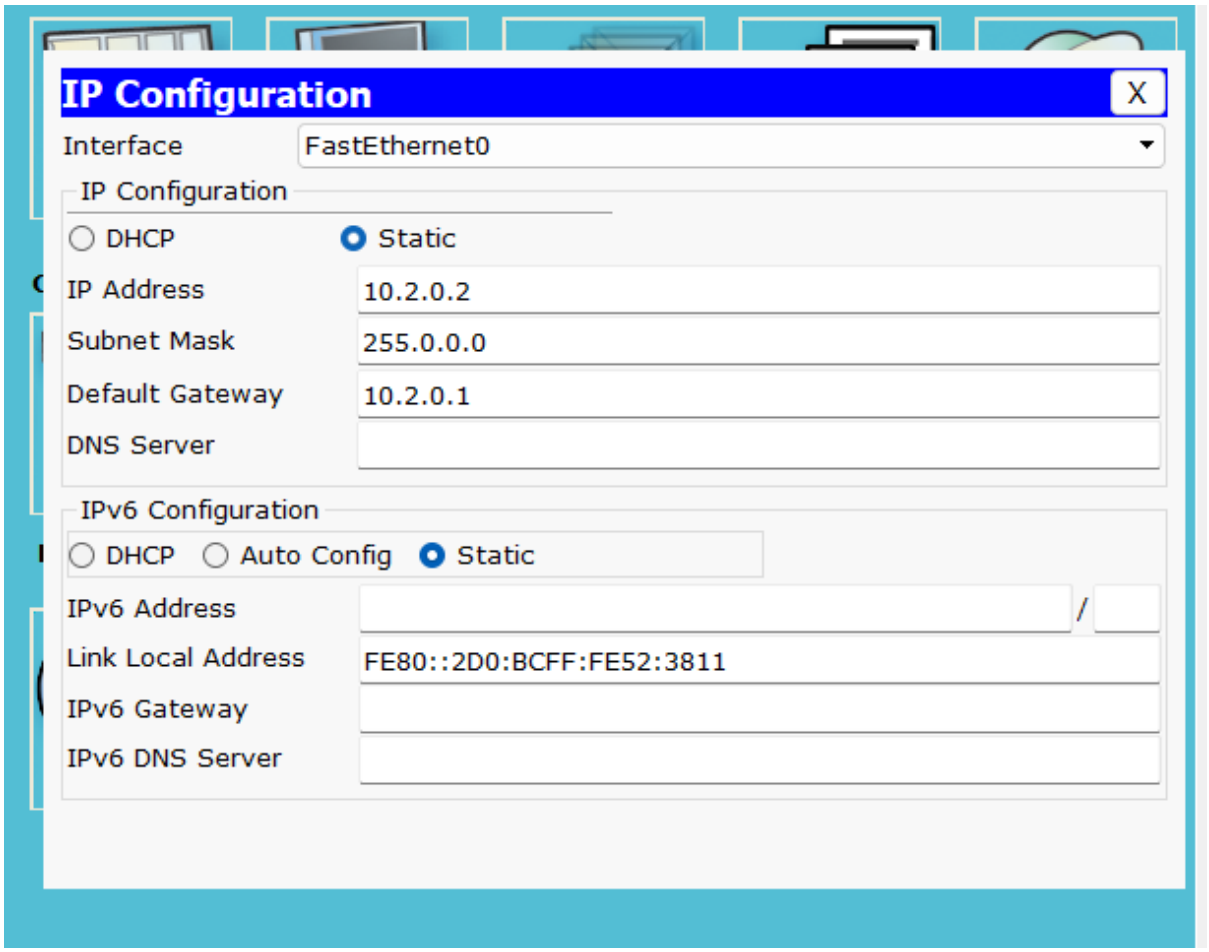
Step 3:

Configure the Server

- Double click on the server to open
- Go to Desktop and add configure the IP

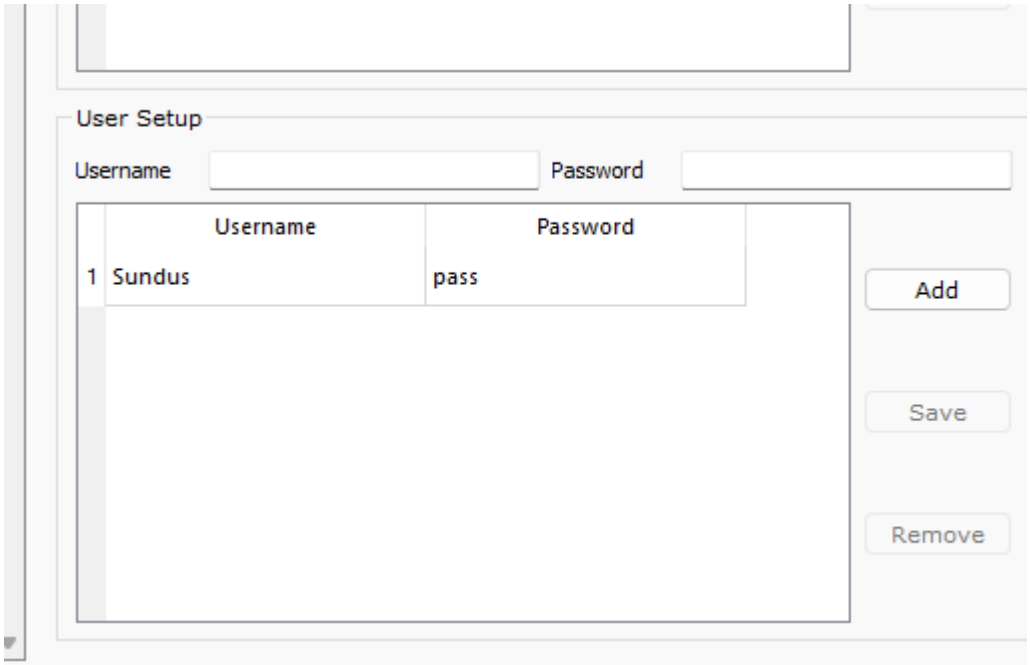
IP address: 10.2.0.2
Subnet Mask: 255.255.255.0
Default Gateway: 10.2.0.1

Insert Screen shot belo



- Go to Services -> AAA make sure it is switched to **on**
  Client Name = vpnServer
  Client IP = 10.2.0.1
  Secret: cisco
  Server Type: Radius
- Click ADD
  Add the Username: roisin (add your own name)
  Add the Password: pass
- Click ADD

Insert Screen shot below:



Close the Server

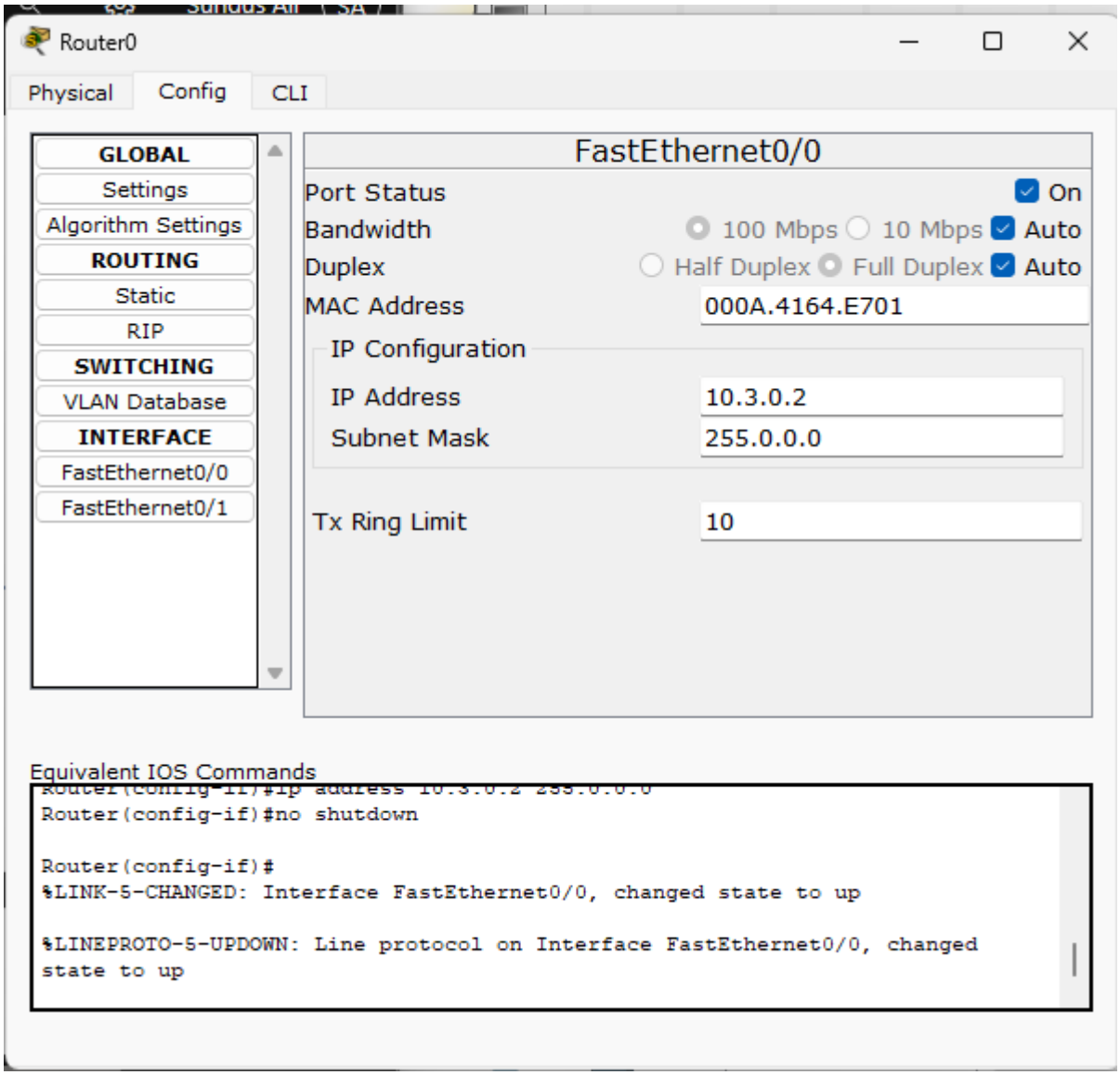
<u>Step 4</u>

Configure the router

- Open Router 0
- Go to config -> interface -> FastEthernet0/0
  Add the IP address 10.3.0.2
  Add the subnet mask 255.255.255.0

  Tick port status check box

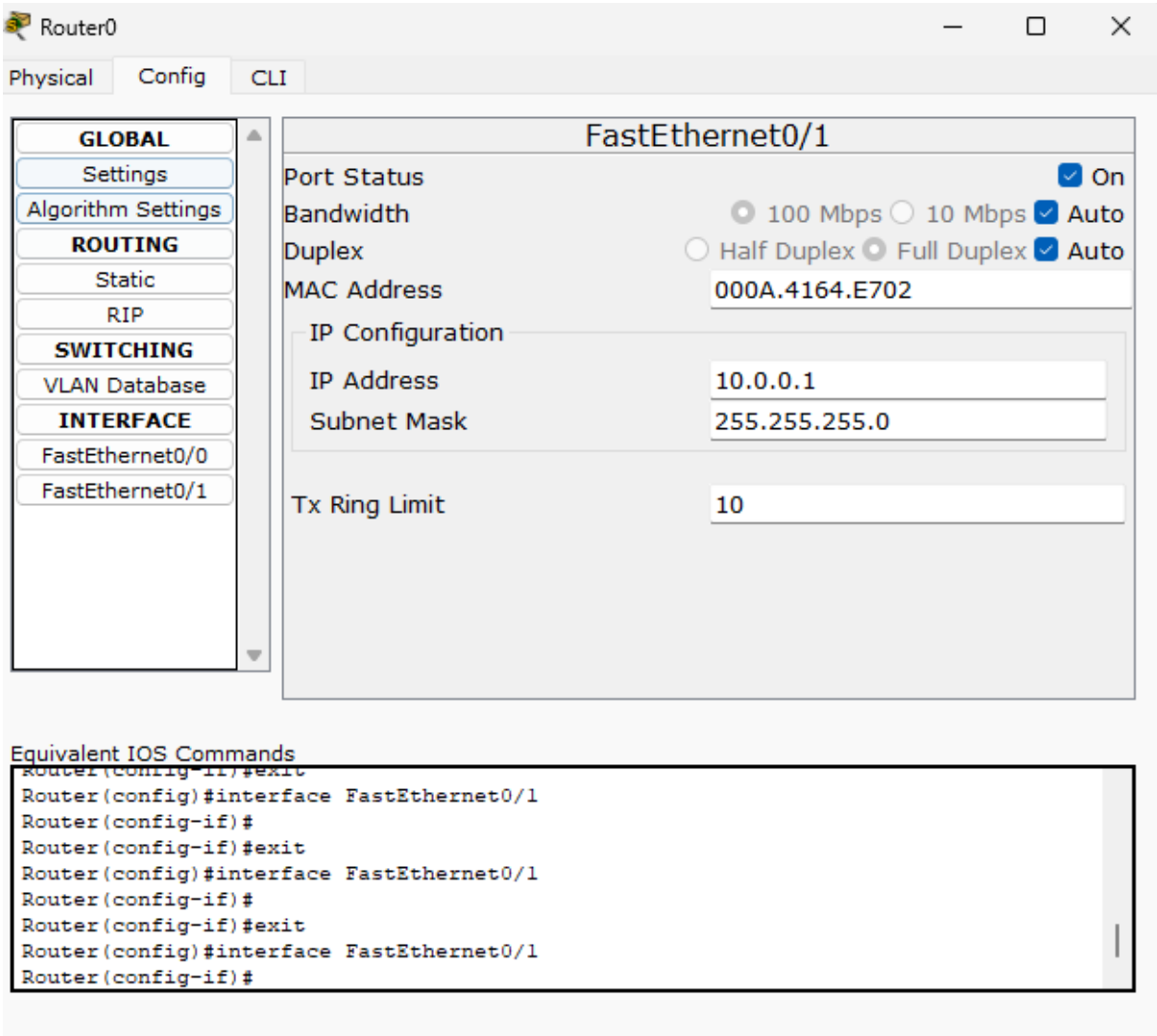
Insert Screenshot below when completed



- Go to config -> interface -> FastEthernet0/1
  Add the IP address 10.0.0.1
  Add the subnet mask 255.255.255.0

  Make sure the port status check box is ticked

Insert Screenshot below when complete



Still in Router0 go to -> Config -> **Routing** -> Static

Add Network: 10.2.0.0
Mask: 255.255.255.0
Next Hop: 10.3.0.1

CLICK Add

Add Network: 10.1.0.0
Mask: 255.255.255.0
Next Hop: 10.3.0.1

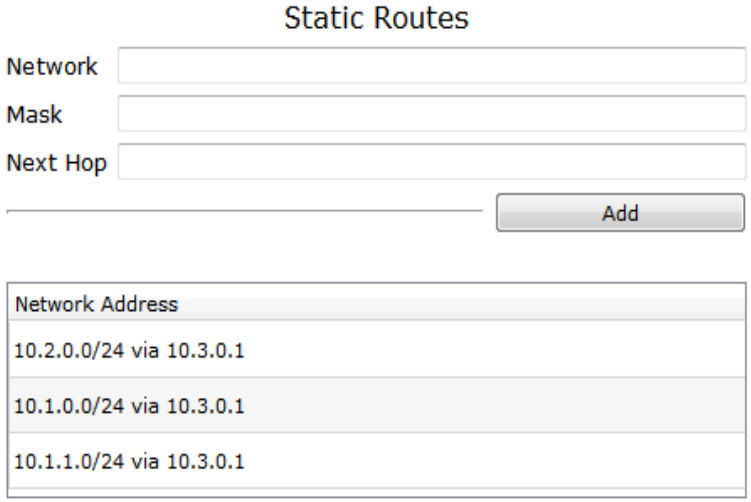
CLICK Add

Add Network: 10.1.1.0

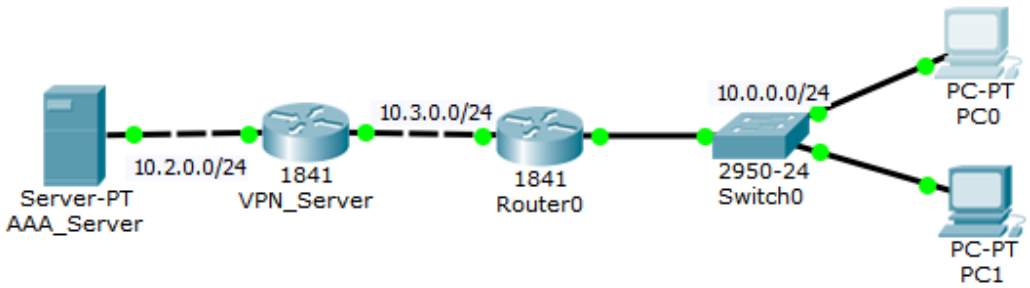
Mask: 255.255.255.0
Next Hop: 10.3.0.1

CLICK Add

Network Address should appear as below



Devices should now be connected (Green lights appear) as shown below



- **Label the interfaces as shown in diagram. Use from your tool bar.**

<u>Step 5</u>

Configure the PCs

- Go to PC0 desktop->IP Configuration
- Add the following details:

Static
IP address: 10.0.0.2
Mask: 255.255.255.0
Gateway 10.0.0.1

- Ping the server ->Go to command prompt
  ping 10.2.0.2

Insert screen shot below:

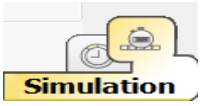
```
Packet Tracer SERVER Command Line 1.0
SERVER>ping 10.2.0.2

Pinging 10.2.0.2 with 32 bytes of data:

Reply from 10.2.0.2: bytes=32 time=10ms TTL=128
Reply from 10.2.0.2: bytes=32 time=1ms TTL=128
Reply from 10.2.0.2: bytes=32 time=1ms TTL=128
Reply from 10.2.0.2: bytes=32 time=11ms TTL=128

Ping statistics for 10.2.0.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 11ms, Average = 5ms

SERVER>
```

- Go to PC1 -> Desktop ->IP Configuration
- Add the following details:
  Static
  IP address: 10.0.0.3
  Mask: 255.255.255.0
  Gateway 10.0.0.1

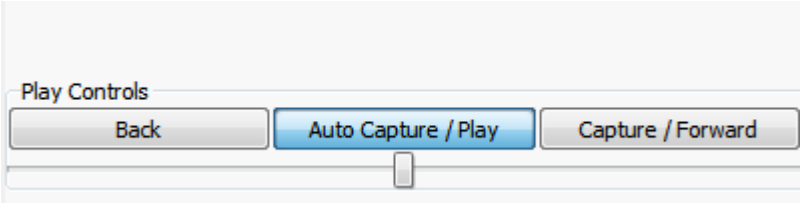
<u>Step 6</u>

Test the connection and routes

- Switch to Simulation Mode



- Go back to PC 0 and ping PC1- Move this window aside
- Press auto capture to show the route the packets take.



What route do the packets take?
It went from pc0 to the switch then from the switch to pc1
And then it came back from pc1 to the switch and from the switch to pc0 and it delivered and got the tick mark.

Insert screen shot below of the simulation panel event list

Time(sec)	Last Device	At Device	Type	Info
901.499	Switch0	PC1	ICMP	
901.500	PC1	Switch0	ICMP	
901.501	Switch0	PC0	ICMP	
902.505	--	PC0	ICMP	
902.506	PC0	Switch0	ICMP	
902.507	Switch0	PC1	ICMP	
902.508	PC1	Switch0	ICMP	
902.509	Switch0	PC0	ICMP	
903.510	--	PC0	ICMP	

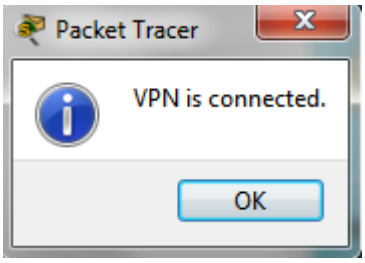
<u>Step 7</u>

VPN connection

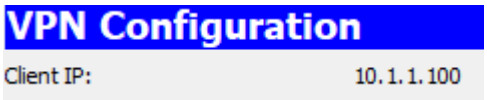
- Switch back to RealTime mode
- Go back to PC0
- Go to VPN configuration

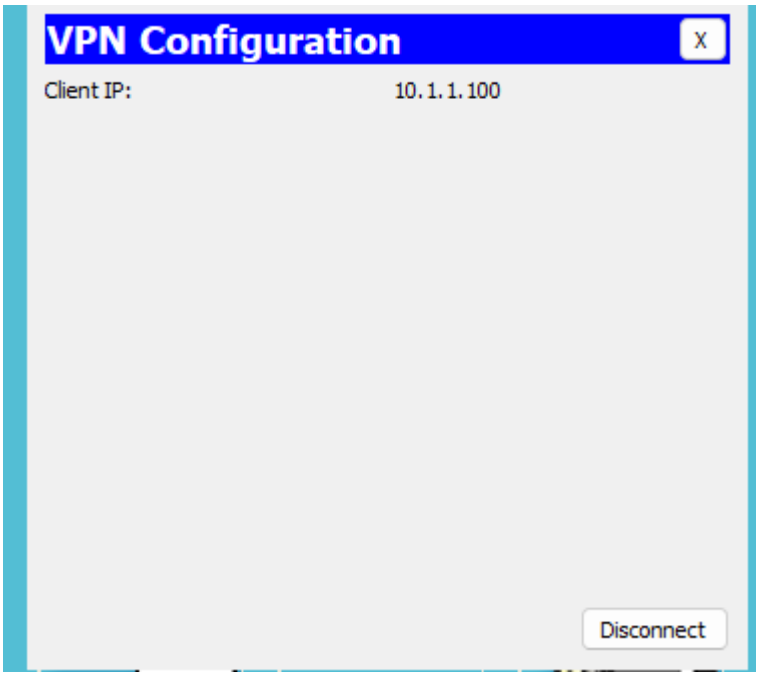
  Add the following information to connect:
  Group name: ciscogroup
  Group Key: ciscogroup
  Host IP(Server IP): 10.3.0.1
  Username: roisin (yourname)
  Password: pass

- Click Connect. The below window should appear



What is the new client IP? Insert screen shot below



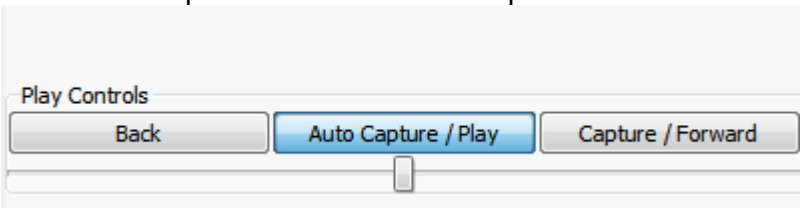


Step 8

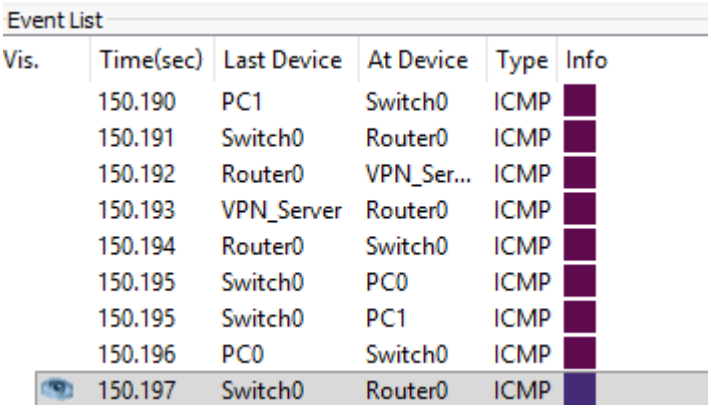
- Go back to PC0 and ping PC1
  Insert screenshot below

```
Packet Tracer PC Command Line 1.0
PC>ping 10.0.0.2

Pinging 10.0.0.2 with 32 bytes of data:

Reply from 10.0.0.2: bytes=32 time=1ms TTL=126
Reply from 10.0.0.2: bytes=32 time=10ms TTL=126
Reply from 10.0.0.2: bytes=32 time=0ms TTL=126
Reply from 10.0.0.2: bytes=32 time=0ms TTL=126

Ping statistics for 10.0.0.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 10ms, Average = 2ms

PC>
```

- In simulation mode ping the PC1 again to see the route the packets take.
- Press Auto Capture to see the route the packets take



Insert screen shot below



Event List

Vis.	Time(sec)	Last Device	At Device	Type	Info
	150.190	PC1	Switch0	ICMP	
	150.191	Switch0	Router0	ICMP	
	150.192	Router0	VPN_Ser...	ICMP	
	150.193	VPN_Server	Router0	ICMP	
	150.194	Router0	Switch0	ICMP	
	150.195	Switch0	PC0	ICMP	
	150.195	Switch0	PC1	ICMP	
	150.196	PC0	Switch0	ICMP	
	150.197	Switch0	Router0	ICMP	

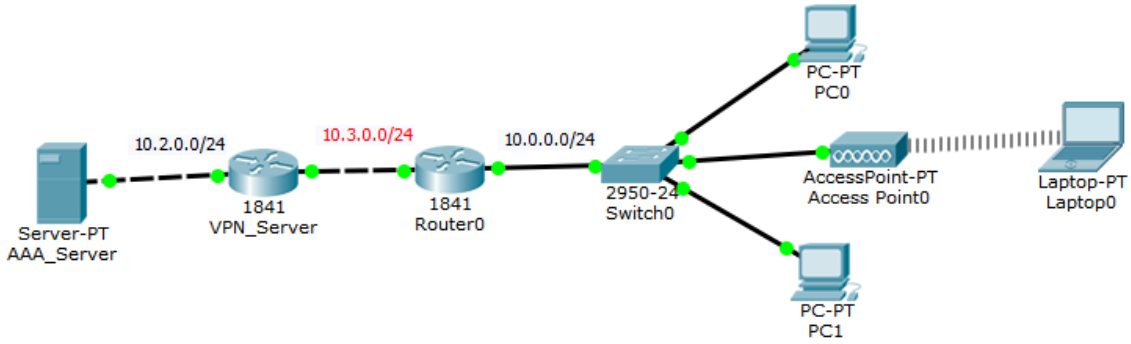
What route do the packets take? Why do you think they do this instead of going straight from the switch?

The packets travel from **PC1 → Switch → Router → VPN** and back to **PC0** because the VPN creates a **secure tunnel** that intercepts and redirects all traffic. Instead of the switch forwarding data directly to a local device, the VPN forces the connection through a remote gateway for encryption. This ensures all data is processed by the VPN server before reaching its final destination, prioritizing security over the shortest physical path.

Step 9

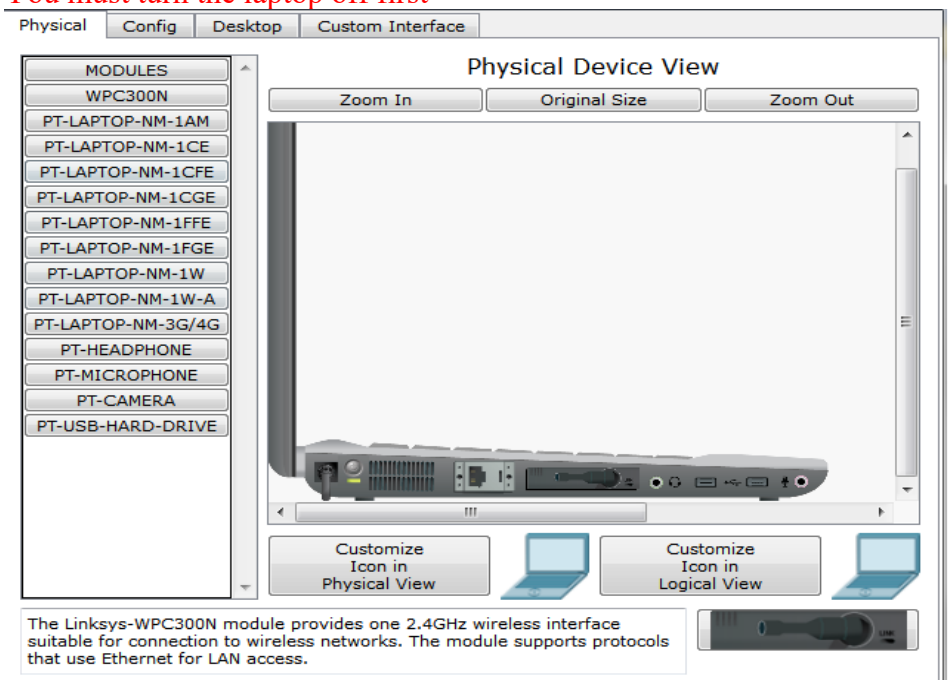
Add a wireless device to network

- Add a wireless access point and a wireless laptop as shown below



- Change the physical connection of the laptop to a Linksy_WPC300N

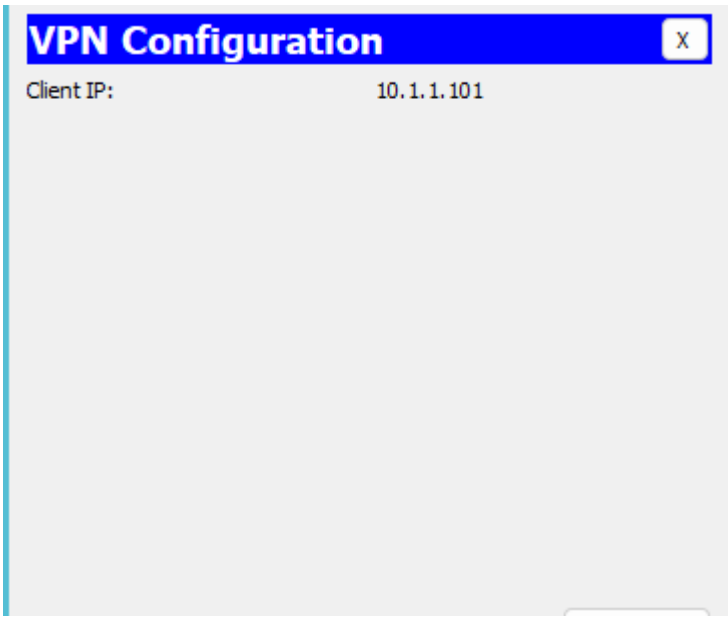
You must turn the laptop off first



- In the Laptop go to Desktop -> IP Configuration
  Add the following:
  Static
  IP address: 10.0.0.4
  Mask: 255.255.255.0
  Gateway 10.0.0.1

- Ping the server 10.2.0.2
- Then connect using VPN
  Group name: ciscogroup
  Group Key: ciscogroup
  Host IP(Server IP): 10.3.0.1
  Username: roisin (yourname)
  Password: pass

What is the client IP:



Answer and explain the following please:

1. **What is a VPN?**

   A VPN (Virtual Private Network) is a service that creates a secure, encrypted connection between your device and the internet. Think of it as a private tunnel for your data to travel through, protecting your information from prying eyes.

2. **Why choose a 'Switch' instead of a 'Hub' for your network?**

   because switches offer significantly better performance, efficiency, and security. Hubs are an outdated, basic technology that simply broadcast all data to every connected device, leading to network congestion and slow speeds.

3. **Outline below how you could test connectivity between a PC in your network and the Server**

   Go to the pc and open the command prompt then Run the Ping: Type ping server address and press Enter.

1. **Success:** You will see "Reply from..." messages.
2. **Failure:** You will see "Request timed out" or "Destination host unreachable".

4. **Outline below how you could test connectivity between a PC in your network and the Router**

   Go to the pc and open the command prompt then Run the Ping: Type ping router address and press Enter.
   Success: You will see "Reply from..." messages.
   Failure: You will see "Request timed out" or "Destination host unreachable".

5. **If there are problems with connectivity, what do you think they would be? Give details of three problems that might occur**
   - ✓ **Wrong IP Address: Mismatched subnets or incorrect default gateways prevent the PC and Server from finding each other logically.**
   - ✓ **Wrong Connection: Using the incorrect cable type or plugging into a port that is "Shut Down" creates a physical link failure (red lights).**
   - ✓ **Bad Network: Configuration errors, such as routing loops or excessive data collisions from using an old hub, cause packets to be dropped**

6. **How would you resolve these problems?**
   1. Fix the IP
   Click PC > Desktop > IP Configuration.
   Manually type in a matching IP Address and Subnet Mask.
   2. Fix the Connection
   Click the Switch or Router > Config tab > Select the Port.
   Check the Port Status "On" box.
   Swap cables: Use the Solid black line for most connections.
   3. Fix the Network
   Click the Fast Forward Time button (Alt+D) to force lights to turn green.
   Click Simulation Mode (Shift+S) and press Play to watch where the packet fails.

   What is your understanding of a network protocol?

A network protocol is a standardized set of rules that determine how data is transmitted and received between devices, ensuring they can communicate even if their hardware or software differs.

What is the TCP/IP?

- **TCP (Transmission Control Protocol) ensures reliability by breaking data into packets, sequencing them, and retransmitting any lost data.**
- **IP (Internet Protocol) handles the logical addressing and routing, making sure packets reach the correct destination across different networks.**

Name and briefly explain the 7 layers of the OSI model?

1. **Application Layer (Layer 7):** The top layer where users interact with network services via software like web browsers or email clients.

2. **Presentation Layer (Layer 6):** Translates, encrypts, and compresses data to ensure it is in a readable format for the application.

3. **Session Layer (Layer 5):** Manages the "conversation" between devices, establishing, maintaining, and terminating communication sessions.

4. **Transport Layer (Layer 4):** Responsible for end-to-end data delivery, error checking, and flow control to ensure complete messages arrive.

5. **Network Layer (Layer 3):** Determines the best physical path for data packets and handles logical addressing (IP addresses) for routing across networks.

6. **Data Link Layer (Layer 2):** Provides reliable node-to-node transfer by breaking packets into frames and using physical addresses (MAC addresses).

7. **Physical Layer (Layer 1):** Transmits raw binary data (bits) over physical media like cables, fiber optics, or wireless signals.